

Gold Technical Indicators

Complete Guide

RSI • SMA/EMA • MACD • Bollinger Bands • Momentum •
Volatility • Support/Resistance

Technical Analysis
Guide

@PPAgoldBot

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1. RSI (Relative Strength Index)

RSI Overview

RSI is an asset momentum indicator "overbought" (overbought) "oversold" (oversold) indicator ranges 0 to 100. Values above 30 indicate oversold (oversold), 70 indicates overbought (overbought).

Formula

RSI uses 14 periods (bot 14) average gain (avg_gain) and average loss (avg_loss) to calculate RS = avg_gain / avg_loss, RSI = 100 - (100 / (1 + RS)). RSI 50 is neutral.

RSI Zones

Zone	RSI Range	Interpretation & Action
Extreme Oversold	0 - 20	Panic selling, buy signal, high risk.
Oversold	20 - 30	buy zone, Bot score: +2 (weight).
Low	30 - 40	Bot score: +1, ideal timing.
Neutral	40 - 60	RSI decision.
High	60 - 70	Bot score: -1, sell signal.
Overbought	70 - 80	Bot score: -2, sell signal.
Extreme Overbought	80 - 100	Correction, sell signal.

RSI Divergence

Bullish Divergence: Price new low, RSI not new low
Price ↓ — RSI ↑ force ↑, price ↑
Price ↓ — RSI ↓ force ↓, price ↓

Bearish Divergence: Price new high, RSI not new high
Price ↑ — RSI ↓ force ↓, price ↓
Price ↓ — RSI ↑ force ↑, price ↑

Limitations

1) **Lagging indicator** — RSI is lagging, price moves first
2) **Strong trend** — Bull market RSI > 70, Bear market RSI < 30
3) **Period 14** — standard balance, noisy, not optimal

Best Practice: RSI decision RSI < 30 (oversold) + Bollinger lower band = buy signal
Bot 5 indicators weighted score combined signal

2. SMA & EMA (Moving Averages)

SMA 简单移动平均

Simple Moving Average 平滑噪音 识别整体趋势 direction 上升 下降 盘整 震荡 盘整 overall trend 上升 SMA 平滑移动平均 SMA5 = 最近 5 个数据点的平均, SMA20 = 20 个数据点的平均

计算公式

$$SMA5 = (Price_1 + Price_2 + Price_3 + Price_4 + Price_5) / 5$$
 简单的公式 计算 5 个数据点的平均

Golden Cross & Death Cross

Signal	Condition	Bot Score & Meaning
Golden Cross	$SMA5 > SMA20$ (短期均线 上穿 长期均线)	Score: -0.5 — 短期均线 上穿 长期均线 看涨信号
Death Cross	$SMA5 < SMA20$ (短期均线 下穿 长期均线)	Score: +1 — 短期均线 下穿 长期均线 看跌信号

Crossover % (ma_crossover_pct)

SMA5 与 SMA20 的交叉百分比 表示 短期均线 与 长期均线 的交叉 0.15% = mild bullish, -0.5% = strong bearish 数字 表示 趋势 的强度

EMA (Exponential Moving Average)

EMA 最近数据 权重 较大 SMA 数据 权重 较小 EMA 平滑移动平均 权重 较大 平滑移动平均 MACD indicator 与 EMA 平滑移动平均 平滑移动平均

Price vs SMA20

Bot 计算 price_vs_sma20 表示 当前价格 与 SMA20 的 差值 与 SMA20 的 百分比 表示 当前价格 与 SMA20 的 差值 与 SMA20 的 百分比 "当前价格 与 SMA20 的 差值 与 SMA20 的 百分比" — 表示 当前价格 与 SMA20 的 差值 与 SMA20 的 百分比

⚠ **Limitations:** 1) Lagging — trend [] SMA
[] 2) Sideways market [] SMA5/SMA20 [] cross
[] false signals [] 3) SMA5 [] noisy, SMA50 [] —
SMA5/SMA20 combo [] hourly data [] reasonable []

3. MACD (Moving Average Convergence Divergence)

MACD 是什么 指标

EMA 指标 converge (收敛) / diverge (发散) 指标
RSI 指标 oversold/overbought 指标, SMA 指标 trend direction 指标
MACD 指标 "momentum 指标"

MACD 指标 3 部分

Component	Formula	What It Shows
MACD Line	EMA12 - EMA26	Short vs Long-term momentum. Positive = bullish, Negative = bearish
Signal Line	MACD Line 9-period EMA	MACD 参考线 (smoothed reference line)
Histogram	MACD Line - Signal Line	! >0 = bullish momentum, <0 = bearish momentum

MACD Crossovers

Bullish Crossover: MACD Line 上穿 Signal Line 指标
— histogram 负值 指标 positive 指标
momentum 指标 指标 signal

Bearish Crossover: 指标 — MACD Line 下穿 Signal Line 指标
momentum 指标 指标

Histogram 指标

Positive + bar 指标 = bullish momentum 指标 Positive + bar 指标 = momentum 指标, 指标 Negative + bar 指标 = bearish momentum 指标 Negative + bar 指标 = 指标 指标, bounce 指标

Zero Line Cross & Divergence

Zero Line Cross: MACD Line > 0 = EMA12 $>$ EMA26, overall bullish trend
 < 0 = bearish trend
Crossover $\uparrow$ $\downarrow$ trend change signal

MACD Divergence: RSI divergence $\uparrow$ $\downarrow$ traders $\uparrow$ $\downarrow$

Bot Scoring

Histogram < 0 : buy_score **+0.5** (bearish = $\downarrow$ = $\downarrow$)

Histogram > 0 : buy_score **-0.5**
MACD weight $\uparrow$ RSI (+/-2) $\downarrow$
 $\downarrow$ — EMA 3 $\downarrow$ lagging $\downarrow$ indicator $\downarrow$
confirmation tool $\downarrow$ $\downarrow$ $\downarrow$

4. Bollinger Bands

Bollinger Bands 是什么 有什么用

Bollinger Bands 由三条线组成，"range" 由 band 3 组成， indicator 由 band 1 和 band 2 组成。 36-37°C Bollinger Bands 由三条线组成，"range" 由 band 3 组成， indicator 由 band 1 和 band 2 组成。 Statistically 95% 在 upper 和 lower band 之间。

Band 3 是什么

Band	Formula	Meaning
Upper Band	$SMA_{20} + (2 \times Std\ Dev)$	Bollinger Bands ceiling
Middle Band	SMA_{20}	Bollinger Bands (20 day SMA)
Lower Band	$SMA_{20} - (2 \times Std\ Dev)$	Bollinger Bands floor

Position % & Bot Scoring

Position	Location	Bot Score
0 - 10%	Lower band 附近	+1.5 (Strong buy signal)
10 - 50%	Lower half	Neutral to slightly favorable
50%	Middle (SMA20)	Neutral
50 - 90%	Upper half	Neutral to slightly unfavorable
90 - 100%	Upper band 附近	-1 (Potential pullback)

Bollinger Squeeze

Bands 变窄 (low volatility) "squeeze" 通常 — Bollinger Bands 变窄，通常预示着 Squeeze 即将发生， band 变窄，通常预示着 big move 即将发生。 Squeeze 之后 breakout 通常 (spring 或 反转)。

Band Walk

Upper band = strong uptrend
Lower band = strong downtrend
Scenario "lower band walk"
rule — lower band walk

△ **Limitations:** 1) Mean reversion — assume — trending market 2) Std Dev 2 — band stay 3) RSI confirm — lower band + RSI < 30 =

5. Momentum (Rate of Change)

Momentum

 speed Formula: (Current Price - Price 10h ago) / Price 10h ago × 100% speedometer
 — direction speed Raw price change
 indicators signal

Momentum Zones

Momentum	Signal	Bot Score
< -0.5%	DECLINING — 	+1 (buy opportunity)
-0.5% to +0.5%	FLAT — 	0 (neutral)
> +0.5%	RISING — 	-0.5 (not ideal to buy)

Momentum Divergence

 new low momentum = " speed
 " — : new high
momentum = " speed " —

⚠ **Limitations:** Raw price change noisy, false signals 10h ago spike " "
 Confirmation tool — RSI < 30 + momentum = strong buy signal

6. Volatility

Volatility

 " " Direction
 — :
 Formula: Std Dev / Mean × 100% (last 20 hours)

Volatility Levels

Volatility	Market State	Implications
~0.01%	Very Calm	Low risk, low opportunity
0.05-0.10%	Normal	
0.20%+	High Volatility	High risk + high opportunity

Key Concepts

Volatility Clustering: Volatile periods are followed by more volatile periods

Bollinger Squeeze = Low Volatility = Big move

Position Sizing: Volatility affects risk (risk is proportional to volatility)

Momentum + Volatility Combined

Momentum	Volatility	Interpretation
Negative	High	Risky but may be oversold
Negative	Low	Urgent exit, avoid
Zero	Very Low	Bollinger Squeeze! Big move, RSI/MACD direction
Positive	High	Profit

7. Support & Resistance

Support/Resistance 是什么 为什么

Support = 某个 "价格暂时企稳" level (暂时性)
Resistance = 某个 "价格暂时受阻" level (暂时性)
indicators 提供 signals 识别 support/resistance 特定 price levels

为什么会有支持/阻力 — Psychology

Support: 某个 level 价格暂时企稳 暂时性 "暂时性" 暂时性, 暂时性 暂时性 \$4,200 暂时性 bounce 暂时性 暂时性 — self-fulfilling prophecy

Resistance: 某个 level 价格暂时受阻 暂时性 "暂时性" 暂时性, 暂时性 暂时性 暂时性

Bot Calculation

Level	Method	Meaning
Support 1	10th percentile (48h)	暂时性 90% 暂时性 暂时性
Support 2	5th percentile (48h)	暂时性 95% 暂时性 floor
Resistance 1	90th percentile (48h)	暂时性 90% 暂时性 暂时性
Resistance 2	95th percentile (48h)	暂时性 95% 暂时性 ceiling

Practical Usage

Buy Decision: 暂时性 Support 1 暂时性 + RSI < 30 = strong buy signal

Target Setting: \$4,180 (support) 暂时性 暂时性 \$4,310 (resistance) 暂时性 profit 暂时性 = \$130/g potential profit

Risk Management: Support 2 暂时性 stop-loss set Risk:Reward ratio 暂时性 trade worth it 暂时性 暂时性

Breakout & Role Reversal

Support     support level 
 resistance  \$4,200 support  \$4,150 
 \$4,200  (breakeven) 
 resistance 

Strength Factors

[illegible]

8. Indicators Combined Strategy

Bot Scoring System

Indicator	Bullish (Buy) Score	Bearish Score
RSI < 30 (oversold)	+2	
RSI 30-40	+1	
RSI 60-70		-1
RSI > 70 (overbought)		-2
SMA5 < SMA20 (death cross)	+1	
SMA5 > SMA20 (golden cross)		-0.5
MACD histogram < 0	+0.5	
MACD histogram > 0		-0.5
Bollinger position < 20%	+1.5	
Bollinger position > 80%		-1
Momentum < -1%	+1	
Momentum > +1%		-0.5

Final Signal (Normalized Score)

Score	Signal	Action
> 1.0	STRONG BUY	Indicators align
0.3 to 1.0	BUY	
-0.3 to 0.3	HOLD	
-1.0 to -0.3	WAIT	

